REVISION NOTES — GCSE AQA PHYSICS: ENERGY

ENERGY — CORE PRINCIPLE

- Energy is the **capacity to do work**.
- Energy is **conserved**: it cannot be created or destroyed, only transferred between stores.
- In real systems, energy becomes **dissipated** (spread to surroundings), usually as thermal energy.
- The total energy before a process = total energy after a process.

ENERGY STORES (AQA-SPECIFIC LIST — MUST MEMORISE)

Energy can be stored in:

- **Thermal store** (internal energy of an object)
- **Kinetic store** (moving objects)
- **Gravitational potential store** (objects raised in a field)
- **Elastic potential store** (stretched or compressed objects)
- **Chemical store** (fuel, food, batteries)
- **Magnetic store**
- **Electrostatic store**
- **Nuclear store**

⚠In AQA exams you must use the phrase **"energy store"**, not "type of energy".

WAYS ENERGY IS TRANSFERRED (EXAM LANGUAGE REQUIRED)

Energy is transferred by:

- **Mechanical work** (force moves an object)
- **Electrical work** (moving charge)
- **Heating** (temperature difference)
- **Radiation** (waves such as light or infrared)

⚠Do NOT say "energy produced". Say **energy transferred**.

WORK DONE (VERY COMMON EXAM TOPIC)

- Work done occurs when a **force causes movement**.
- Work done = energy transferred.

Formula:

$W = F \times d$

Where:

- W = work done (J)
- F = force (N)
- d = distance moved in direction of force (m)

KINETIC ENERGY (MOVING OBJECTS)

Formula:

$E_k = \frac{1}{2} m v^2$

Where:

- m = mass (kg)
- v = speed (m/s)

Key points:

- Speed is squared → small speed increase gives large energy increase.
- Used in braking distance questions.

GRAVITATIONAL POTENTIAL ENERGY (RAISED OBJECTS)

Formula:

$E_p = m g h$

Where:

- m = mass (kg)
- g = gravitational field strength (9.8 N/kg unless told otherwise)
- h = height (m)

Used in:

- Falling objects
- Rollercoaster
- Lifting loads

ELASTIC POTENTIAL ENERGY (SPRINGS / STRETCHING)

Formula:

$Ee = ½ k e^2$

Where:

- k = spring constant (N/m)
- e = extension (m)

Applies only if object obeys **Hooke's Law**.

CONSERVATION OF ENERGY IN SYSTEMS

Example chain:

Chemical store (battery)
→ electrical transfer
→ kinetic store (motor rotation)
→ thermal store (wasted heat)

Exam tip: always show **full energy pathway**.

DISSIPATION OF ENERGY

- Energy is not destroyed but becomes **less useful**.
- Usually spreads into surroundings as thermal energy.
- This is called **dissipation**.

Example:
Car braking → kinetic store → thermal store of brakes + surroundings

⚠Never write "energy lost". Write **energy dissipated**.

EFFICIENCY (HIGH-MARK TOPIC)

Efficiency shows how much input energy becomes useful output.

Formula:

Efficiency = useful output energy / total input energy

OR

Efficiency = useful power output / total power input

Percentage version:

Efficiency (%) = (useful output ÷ total input) × 100

Key exam points:

- Efficiency always ≤ 1 (or ≤ 100%)
- Improve efficiency by reducing unwanted energy transfers
- Insulation improves efficiency

POWER (RATE OF ENERGY TRANSFER)

Definition:
Power = how quickly energy is transferred.

Formula:

$P = E / t$

Also:

$P = W / t$

Where:

- P = power (W)
- E = energy (J)
- t = time (s)

ENERGY RESOURCES (AQA CLASSIFICATION)

Renewable:

- Solar
- Wind
- Hydro
- Geothermal
- Tidal
- Waves
- Bio-fuel

Non-renewable:

- Coal
- Oil
- Gas
- Nuclear fuel

Exam answers often require:

✔ Reliability
✔ environmental impact
✔ startup time
✔ cost

REQUIRED PRACTICAL — THERMAL INSULATION

Typical method:

1. Fill containers with hot water
2. Wrap each with different insulating material
3. Measure temperature drop over time
4. Compare cooling rates

Best insulator = **smallest temperature decrease**

COMMON GCSE ENERGY EXAM ERRORS (VERY IMPORTANT)

Students often:

- Write “energy used up” instead of **transferred**
- Forget to square velocity in kinetic energy

- Use cm instead of meters in formulas
- Forget units in answers
- Say “force = work done” (incorrect)
- Say “energy created” (physically wrong)

EXAMINER KEYWORDS THAT SCORE MARKS

Use these exact phrases when possible:

- “energy transferred mechanically”
- “energy dissipated to the surroundings”
- “conservation of energy”
- “useful energy output”
- “thermal store increases”